Sunday, 3 May 2026 6:10 pm

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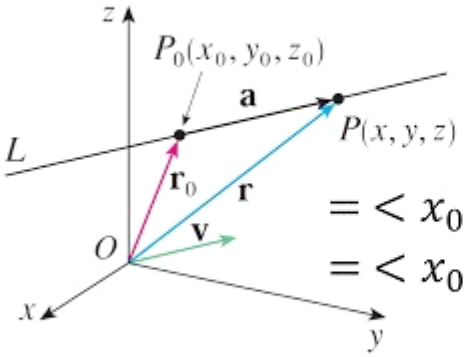
Parametric Equations of a Line



$$\boldsymbol{r} = \boldsymbol{r}_0 + t\boldsymbol{v}$$
$$= < x, y, z >$$
$$= < x_0, y_0, z_0 > + t < a, b, c >$$
$$= < x_0 + ta, y_0 + tb, z_0 + tc >$$

$$\boxed{x = x_0 + at \quad y = y_0 + bt \quad z = z_0 + ct}$$

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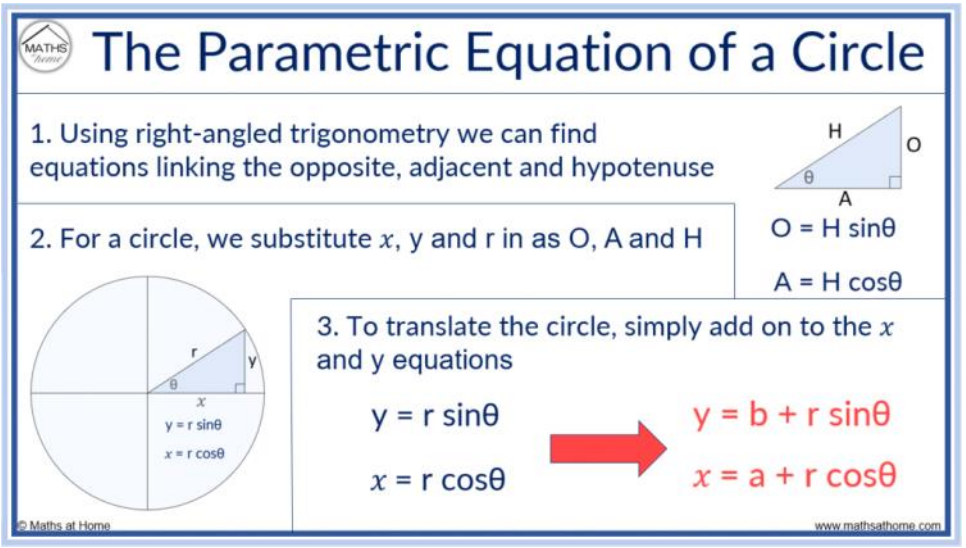


Parametric Surfaces

A parameterized surface is given by a description of the form

$$\mathbf{r}(u, v) = \langle x(u, v), y(u, v), z(u, v) \rangle$$

Describe surface S parameterized by

$$\mathbf{r}(u, v) = \langle \cos u, \sin u, v \rangle, -\infty < u < \infty, -\infty < v < \infty.$$

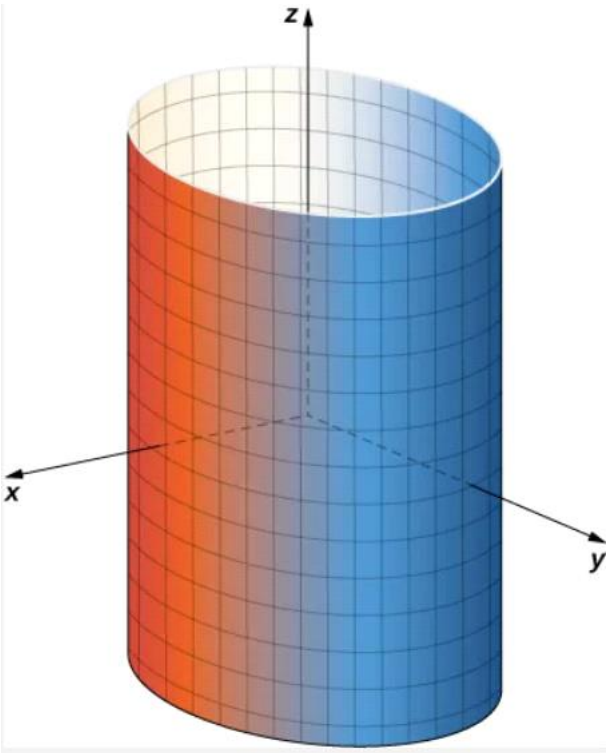
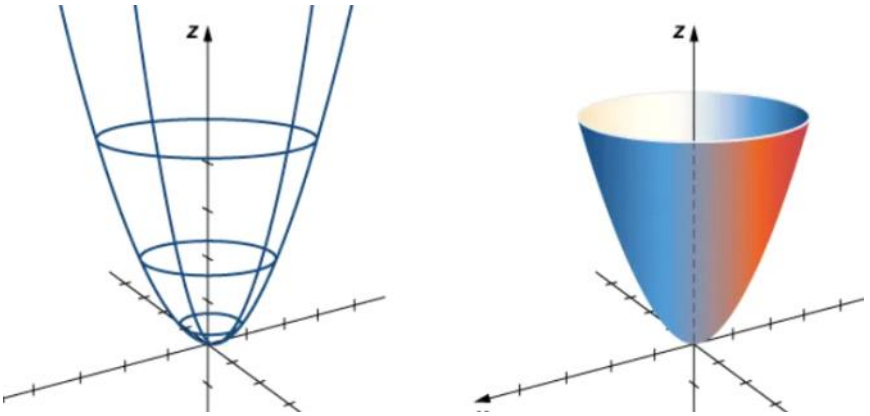


Figure 6.58 Cylinder $x^2 + y^2 = r^2$ has parameterization $\mathbf{r}(u, v) = \langle r\cos u, r\sin u, v\rangle$, $0 \le u \le 2\pi, -\infty < v < \infty$.

Describe surface S parameterized by

$$\mathbf{r}(u, v) = \left\langle u\cos v, u\sin v, u^2\right\rangle, 0 \le u < \infty, 0 \le v < 2\pi.$$



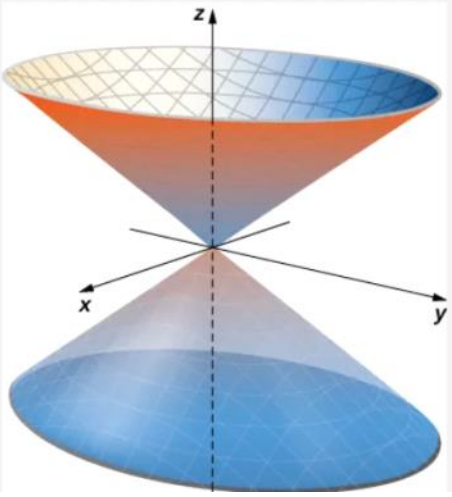
$$\begin{aligned} x^2 + y^2 &= (u\cos v)^2 + (u\sin v)^2 \\ &= u^2\cos^2 v + u^2\sin^2 v \\ &= u^2 \\ &= z. \end{aligned}$$

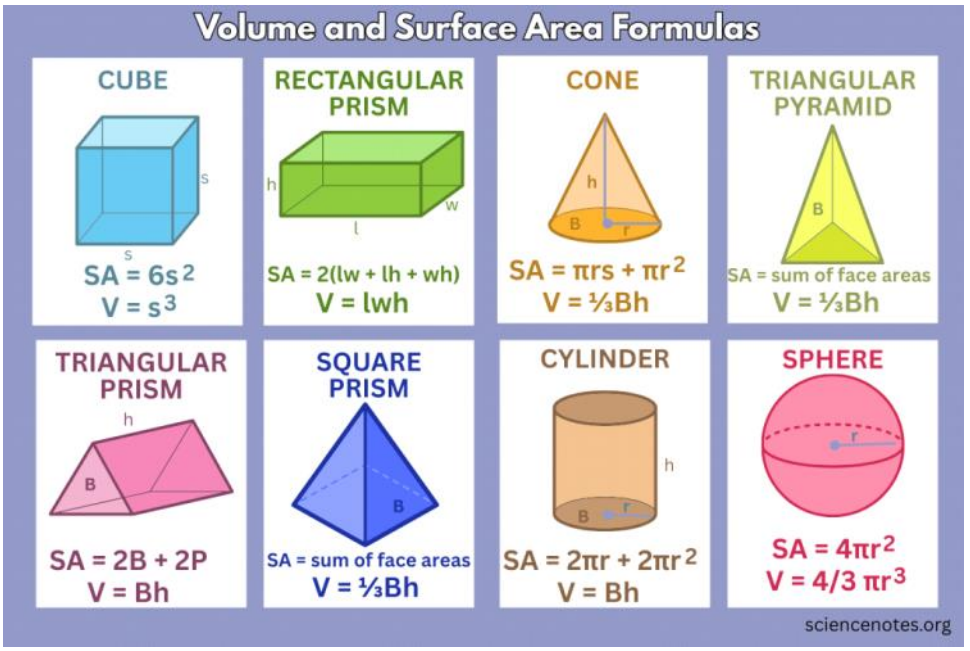
Finding a Parameterization

Give a parameterization of the cone $x^2 + y^2 = z^2$ lying on or above the plane $z = -2$.

Solution

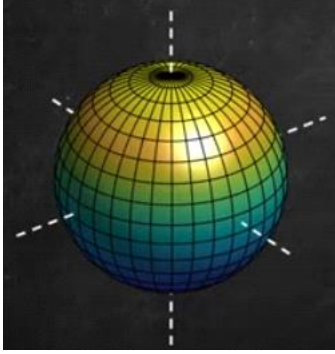
The horizontal cross-section of the cone at height $z = u$ is circle $x^2 + y^2 = u^2$. Therefore, a point on the cone at height u has coordinates $(u \cos v, u \sin v, u)$ for angle v. Hence, a parameterization of the cone is $\mathbf{r}(u, v) = \langle u \cos v, u \sin v, u \rangle$. Since we are not interested in the entire cone, only the portion on or above plane $z = -2$, the parameter domain is given by $-2 \leq u < \infty, 0 \leq v < 2\pi$ (Figure 6.60).





A smooth surface S has surface area

$$\iint_S d\sigma$$



$$SA = \iint_R |\vec{r}_u \times \vec{r}_v|\, dudv$$

Implicit: $F(x, y, z) = c$

$$SA = \iint_R \frac{|\nabla F|}{|\nabla F \cdot \vec{p}|} dA$$

Explicit: $z = f(x, y)$

$$SA = \iint_R \sqrt{f_x^2 + f_y^2 + 1}\, dA$$

$$Surface\ Integral = \iint_S G(x, y, z) d\sigma$$

$$SI = \iint_R G(f(u, v), g(u, v), h(u, v)) |\vec{r}_u \times \vec{r}_v|\, dudv$$

Implicit: $F(x,y,z) = c$

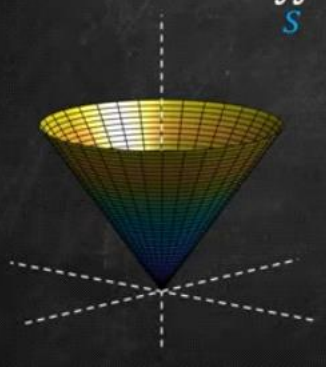
$$SI = \iint_R G(x,y,z)\frac{|\nabla F|}{|\nabla F \cdot \vec{p}|}\,dA$$

Explicit: $z = f(x,y)$

$$SI = \iint_R G(x,y,f(x,y))\sqrt{f_x^2 + f_y^2 + 1}\,dA$$

Let $\delta(x,y,z)$ be the mass density of a thin shell S

$$Total\ Mass = \iint_S \delta(x,y,z)d\sigma$$



Let $T(\phi,\theta) = -20 + 50\sin\phi$ be the temperature on a sphere of radius a.

$$Average\ Temp = \frac{1}{4\pi a^2}\iint_S T(\phi,\theta)d\sigma$$

$$= \frac{1}{4\pi a^2}\int_0^{2\pi}\int_0^{\pi}(-20 + 50\sin\phi)\underbrace{a^2\sin\phi}_{|\vec{r}_\phi \times \vec{r}_\theta|}\,d\phi d\theta$$

$$= \frac{a^2}{4\pi a^2} \int_0^{2\pi} \int_0^{\pi} \left(-20 \sin\phi + \frac{50}{2}(1 - \cos(2\phi)) \right) d\phi d\theta$$

$$= \frac{2\pi a^2 (25\pi - 40)}{4\pi a^2} \approx 18° \text{ C}$$

First, let's look at the surface integral in which the surface S is given by $z = g(x, y)$. In this case the surface integral is,

$$\iint_S f(x, y, z)\, dS = \iint_D f(x, y, g(x, y)) \sqrt{\left(\frac{\partial g}{\partial x}\right)^2 + \left(\frac{\partial g}{\partial y}\right)^2 + 1}\, dA$$

The second method for evaluating a surface integral is for those surfaces that are given by the parameterization,

$$\vec{r}(u, v) = x(u, v)\vec{i} + y(u, v)\vec{j} + z(u, v)\vec{k}$$

In these cases the surface integral is,

$$\iint_S f(x, y, z)\, dS = \iint_D f(\vec{r}(u, v)) \|\vec{r}_u \times \vec{r}_v\|\, dA$$

where D is the range of the parameters that trace out the surface S.

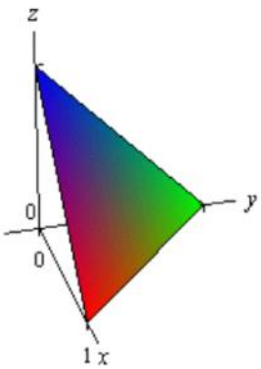
$$\vec{r}(x, y) = x\vec{i} + y\vec{j} + g(x, y)\vec{k}$$

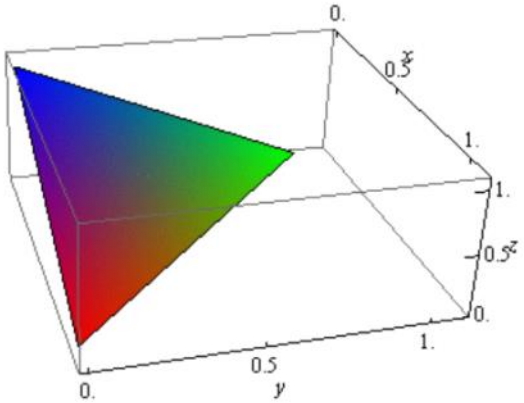
$$\|\vec{r}_x \times \vec{r}_y\| = \sqrt{\left(\frac{\partial g}{\partial x}\right)^2 + \left(\frac{\partial g}{\partial y}\right)^2 + 1}$$

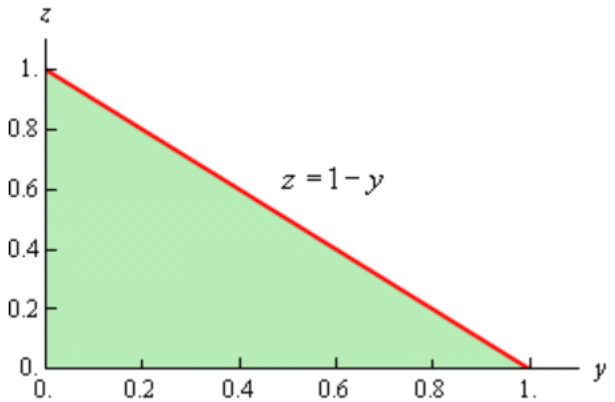
Example 1 Evaluate $\iint_S 6xy\, dS$ where S is the portion of the plane $x + y + z = 1$ that lies in the 1st octant and is in front of the yz-plane.

$$x = 1 - y - z$$

ıst what D is. Here is a sketch of the surface S.







$$0 \le y \le 1 \qquad 0 \le z \le 1 - y$$

$$\iint_S 6xy\,dS = \iint_D 6\,(1 - y - z)\,y\sqrt{1 + (-1)^2 + (-1)^2}\,dA$$

$$\begin{aligned}\iint_S 6xy\,dS &= \sqrt{3}\iint_D 6\left(y-y^2-zy\right)\,dA\\ &= 6\sqrt{3}\int_0^1\int_0^{1-y} y-y^2-zy\,dz\,dy\\ &= 6\sqrt{3}\int_0^1 \left(yz-zy^2-\frac{1}{2}z^2y\right)\Big|_0^{1-y}\,dy\\ &= 6\sqrt{3}\int_0^1 \frac{1}{2}y-y^2+\frac{1}{2}y^3\,dy\\ &= 6\sqrt{3}\left(\frac{1}{4}y^2-\frac{1}{3}y^3+\frac{1}{8}y^4\right)\Big|_0^1 = \frac{\sqrt{3}}{4}\end{aligned}$$

Example 3 Evaluate $\iint_S y\,dS$ where S is the portion of the cylinder $x^2+y^2=3$ that lies between $z=0$ and $z=6$.

We parameterized up a cylinder in the previous **section**. Here is the parameterization of this cylinder.

$$\vec{r}\left(z,\theta\right)=\sqrt{3}\cos\theta\,\vec{i}+\sqrt{3}\sin\theta\,\vec{j}+z\,\vec{k}$$

The ranges of the parameters are,

$$0\le z\le 6 \qquad 0\le\theta\le 2\pi$$

Now we need $\vec{r}_z\times\vec{r}_\theta$. Here are the two vectors.

$$\begin{aligned}\vec{r}_z\left(z,\theta\right)&=\vec{k}\\ \vec{r}_\theta\left(z,\theta\right)&=-\sqrt{3}\sin\theta\,\vec{i}+\sqrt{3}\cos\theta\,\vec{j}\end{aligned}$$

Here is the cross product.

$$\begin{aligned}\vec{r}_z\times\vec{r}_\theta &= \begin{vmatrix}\vec{i} & \vec{j} & \vec{k}\\ 0 & 0 & 1\\ -\sqrt{3}\sin\theta & \sqrt{3}\cos\theta & 0\end{vmatrix}\\ &= -\sqrt{3}\cos\theta\,\vec{i}-\sqrt{3}\sin\theta\,\vec{j}\end{aligned}$$

$$\|\vec{r}_z \times \vec{r}_\theta\| = \sqrt{3\cos^2\theta + 3\sin^2\theta} = \sqrt{3}$$

$$\begin{aligned}
\iint_S y\,dS &= \iint_D \sqrt{3}\sin\theta\left(\sqrt{3}\right)\,dA \\
&= 3\int_0^{2\pi}\int_0^6 \sin\theta\,dz\,d\theta \\
&= 3\int_0^{2\pi} 6\sin\theta\,d\theta \\
&= (-18\cos\theta)\big|_0^{2\pi} \\
&= 0
\end{aligned}$$